

SigLIP2 for aerial fire risk classification

Authors to be confirmed

Abstract

We examine the transfer of a pretrained SigLIP2 image encoder to seven class fire risk classification from aerial imagery. We introduce a reproducible partition of the public FireRisk training mirror and an implementation that records data provenance, preprocessing and model selection. Two initial runs compare a frozen encoder probe with full model adaptation. On the validation partition, full adaptation reaches 63.05% accuracy and 58.94% macro F1, compared with 55.95% and 50.19% for the probe. Both runs use one training seed and select their checkpoint on the same validation partition. These development results support further evaluation of SigLIP2 but do not establish performance on an independent test set or unseen regions. The accompanying code provides a common framework for repeated experiments and comparisons with additional visual encoders.

1 Introduction

Aerial images describe vegetation, buildings and surface cover that may help distinguish wildfire hazard categories. FireRisk links such imagery to labels derived from the Wildfire Hazard Potential map [1]. The task is to recover a hazard category assigned to an image tile. It does not directly measure the probability of a future fire at that location.

SigLIP2 learns visual representations using image and text supervision together with additional representation learning objectives [2]. This motivates an empirical question about whether its image encoder can adapt to the different visual content of aerial imagery. We study a small initial comparison and release its protocol and implementation for further experiments.

2 Data and protocol

The original FireRisk study describes 91,872 images, with 70,331 training examples and a separate validation partition [1]. Its RGB imagery was obtained from the National Agriculture Imagery Program during 2019 and 2020 and cropped to 320×320 pixels. Labels come from the 2020 Wildfire Hazard Potential map. We use the `blanchon/FireRisk` mirror, which exposes only the 70,331 training examples. The mirror therefore does not provide the published evaluation partition.

We create a new split with 49,231 training images, 10,552 validation images and 10,548 reserved test images. Allocation is stratified by class, using split seed 2026. The seven categories are very low, low, moderate, high, very high, nonburnable and water. The split preserves the class imbalance. For example, the validation partition contains 3,264 very low examples and 490 very high examples.

Before allocation, the preparation code hashes decoded RGB pixels and image dimensions using SHA256 after correcting image orientation. It groups exact duplicates and rejects conflicting labels within a group. The audit found 70,331 unique pixel groups and no repeated images in this revision. This check concerns exact identity. It cannot exclude overlapping tiles or

visual similarity between nearby locations. The mirror contains image and label fields without coordinates or acquisition dates, so this experiment cannot establish geographic or temporal independence.

The data revision, model revision and split hash are recorded with each run. Validation macro F1 selects the checkpoint. The reserved test partition has not been evaluated in this study. As a result, our scores are development measurements under a new protocol and cannot be compared directly with the original FireRisk benchmark.

3 Model and training

We use the SigLIP2 base image encoder with 16×16 patches at 224×224 input resolution, distributed through timm. The pretrained attention pooling is retained. A trainable LayerNorm and a seven output linear layer operate on its 768 dimensional representation. The resulting classifier has 92,891,143 parameters. In the frozen encoder probe, only the head changes, leaving 6,919 trainable parameters. The encoder remains in evaluation mode during training. Full adaptation updates all parameters and uses head dropout with probability 0.1.

Training uses random crops covering 80% to 100% of the image area, with aspect ratios from 0.9 to 1.1. Horizontal and vertical flips have probability 0.5, and rotations use multiples of 90 degrees. Evaluation uses deterministic bicubic resizing and the pretrained center crop policy with crop proportion 0.9. Each RGB channel is normalized with mean 0.5 and standard deviation 0.5.

Both runs use AdamW with weight decay 0.05, label smoothing 0.05 and gradient clipping at norm 1. Learning rates follow a cosine schedule after warmup over 10% of the configured optimization steps. The probe uses a head learning rate of 10^{-3} and at most 15 epochs. Full adaptation uses 10^{-5} for the encoder and 3×10^{-4} for the head, with at most 30 epochs. No class weighting or mixup is applied. The effective batch size is 64 in both cases. Full adaptation accumulates gradients over two batches of 32 images and uses gradient checkpointing. Computation uses bfloat16 on one NVIDIA RTX 5090 GPU. Training seed 42 is shared by the two runs.

Early stopping follows seven epochs without a validation macro F1 improvement greater than 0.0001. Both runs stop after 14 epochs and select epoch 7. The different learning rates, dropout and schedule lengths make this a comparison of two initial recipes. They prevent attribution of the observed difference to encoder adaptation alone.

4 Preliminary results

Table 1 reports uncalibrated predictions from the selected checkpoints. Full adaptation improves validation accuracy by 7.10 percentage points and macro F1 by 8.75 percentage points relative to the probe. Macro F1 assigns equal weight to the seven class F1 scores, making the minority classes visible alongside accuracy.

Recipe	Trainable parameters	Accuracy (%)	Macro F1 (%)	Time (min)	Memory (GiB)
Frozen encoder probe	6,919	55.95	50.19	12.1	0.74
Full adaptation	92,891,143	63.05	58.94	19.0	1.96

Table 1. Validation results from one training seed. Time covers the training loop, validation and checkpoint writing, excluding data preparation and model loading. Memory is peak PyTorch allocated GPU memory and does not represent total device or system memory.

The strongest class F1 scores after full adaptation are water at 87.33% and nonburnable at 85.12%. Very high reaches 39.90% F1 and 33.47% recall, compared with 21.29% F1 and 14.49% recall for the probe. Although this minority category improves, its recall remains low. Full adaptation correctly identifies 164 of its 490 validation examples. This limits the usefulness of the current classifier when missing high hazard areas has a substantial cost.

The runs executed concurrently on separate RTX 5090 GPUs. Their timings are contextual measurements and do not provide a controlled speed comparison. Times exclude environment installation, downloads and the initial pixel audit. No repeated training seeds are available, so the variation caused by training randomness remains unknown.

5 Reproducibility and limitations

The implementation keeps original Parquet shards and can build an indexed archive of unchanged image bytes for shuffled access. It avoids extracting each image to a separate file and records the selected artifact location. Each run saves its configuration, environment, preprocessing, data manifest, training history and selected weights. Immutable dataset and model revisions and the split hash are listed in the accompanying manuscript documentation. The initial run artifacts do not record a source commit and mark the development tree as modified. The repository also includes presets for LoRA adaptation, DINOv2, ConvNeXt and ResNet50. Those presets have no completed comparative results here.

Several steps remain before a broader empirical claim is warranted. The recipes should receive declared validation search budgets and be repeated with independent training seeds. The fixed recipes can then be evaluated on the reserved test partition. A stronger study of transfer to new locations requires geographic metadata and a separate spatial holdout. Exact pixel hashes do not resolve geographic correlation or possible overlap with the pretraining data.

Full SigLIP2 adaptation merits further study on this mirror. Errors within the risk categories, particularly very high hazard, remain a central concern.

Code availability github.com/yunusserhat/firerisk. The code is distributed under the GNU General Public License version 3. Upstream imagery and pretrained weights retain their respective terms.

References

- [1] Shuchang Shen, Sachith Seneviratne, Xinye Wanyan, and Michael Kirley. FireRisk: A remote sensing dataset for fire risk assessment with benchmarks using supervised and self-supervised learning, 2023. URL <https://arxiv.org/abs/2303.07035>.
- [2] Michael Tschannen, Alexey Gritsenko, Xiao Wang, Muhammad Ferjad Naeem, Ibrahim Alabdulmohsin, Nikhil Parthasarathy, Talfan Evans, Lucas Beyer, Ye Xia, Basil Mustafa, Olivier Hénaff, Jeremiah Harmsen, Andreas Steiner, and Xiaohua Zhai. SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features, 2025. URL <https://arxiv.org/abs/2502.14786>.